

Radical Axes

Problem 1. Let $ABCD$ be a cyclic quadrilateral. The internal angle bisector of $\angle BAC$ meets BC at E . The internal angle bisector of $\angle BDC$ meets BC at F . The circumcircles of $\triangle ABE$ and $\triangle CDF$ meet at two distinct points P and Q . Prove that AE , DF and PQ are concurrent.

Problem 2. In $\triangle ABC$, the points D and E lie on the sides AB and AC respectively such that $AD \times DB = AE \times EC$. The circumcircles of $\triangle ABC$ and $\triangle ADE$ meet again at F . Prove that $AF \parallel DE$.

Problem 3. Given $\triangle ABC$, two variable points X and Y lie on the sides AB and AC respectively. Suppose the circles with diameters BY and CX respectively intersect at two distinct points P and Q . Prove that the line PQ passes through a fixed point when X and Y vary.

Problem 4. Let $ABCD$ be a cyclic quadrilateral. The diagonals AC and BD meet at E . The extension of the lines AD and BC meet at F . Let H_1 and H_2 be the orthocentres of $\triangle ABF$ and $\triangle CDF$ respectively. Prove that E , H_1 and H_2 are collinear.

Problem 5. In $\triangle ABC$, let D, E, F be the feet of the altitudes from A, B, C respectively. The lines BC and EF meet at X . The lines CA and FD meet at Y . The lines AB and DE meet at Z . Prove that X, Y, Z lie on a line perpendicular to the Euler line of $\triangle ABC$.

Problem 6. In $\triangle ABC$, let D and E be two points on the side BC . Let P and Q be points on the sides AB and AC respectively. The lines DP and EQ meet at X . Prove that X lies on the radical axis of the circumcircles of $\triangle ABE$ and $\triangle ACD$ if and only if $PQ \parallel BC$.

Problem 7. Acute triangle ABC is inscribed in circle ω . Let H and O denote its orthocentre and circumcentre, respectively. Let M and N be the midpoints of sides AB and AC , respectively. Rays MH and NH meet ω at P and Q , respectively. Lines MN and PQ meet at R . Prove that $OA \perp RA$.

Problem 8. The incircle of $\triangle ABC$ touches the sides BC, CA, AB at points D, E, F respectively. Let ℓ_1 be the line equidistant to point A and the line EF , ℓ_2 the line equidistant to point B and the line FD , and ℓ_3 the line equidistant to point C and the line DE . Prove that the circumcentre of the triangle bounded by ℓ_1, ℓ_2 and ℓ_3 coincides with the circumcentre of $\triangle ABC$.

Problem 9. Let D be a point on the side BC of $\triangle ABC$ such that $\angle CAD = \angle CBA$. Circle O passing through the points B, D intersects the segments AB, AD at E, F respectively. BF and DE intersect at G . M is the midpoint of AG . Prove that $CM \perp AO$.

Problem 10. In $\triangle ABC$, the internal angle bisector of $\angle BAC$ meets the side BC at D . The common external tangent of the circumcircles of $\triangle ABD$ and $\triangle ACD$ which is closer to D touches the two circles at E and F respectively. Let P be the point on the ray AD such that $\triangle ABP \sim \triangle APC$. Prove that $DEPF$ is a parallelogram.